HAFNet: Hierarchical Attention Fusion Network for Infrared Small Target Detection

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***Abstract*—Infrared small target detection (IRSTD) involves identifying targets that are typically small in spatial extent, have low signal-to-clutter ratios, and are often embedded in dynamic and complex backgrounds, making the task particularly challenging. Benefiting from the powerful feature extraction and multiscale feature fusion characteristics, U-Net performs well in the IRSTD task. However, existing U-Net methods often focus solely on optimizing backbone feature extraction or skip connections, which limits their performance in complex scenes and makes it difficult to recognize small targets effectively. To address this limitation, we propose a novel hierarchical attention fusion network based on the U-Net architecture, namely HAFNet. Specifically, a dual-branch semantic perception module (DSPM) is designed as the feature extraction backbone to enhance contextual semantic interactions. This module integrates dual-branch feature extraction using standard and dilated convolutions while utilizing spatial and channel attention modules (CAMs) to effectively separate small targets from background noise. In addition, we extend the skip connection by merging a hierarchical feature fusion encoder (HFFE) and a hierarchical feature fusion decoder (HFFD). These modules utilize hierarchical attention-guided and encoded feature injection skip connections (ESCs) to achieve effective fusion of multiscale and multilevel semantic features between the encoder and decoder. Extensive experiments on three public datasets (NUAA-SIRST, IRSTD-1K, and NUDT-SIRST) demonstrate that the proposed HAFNet outperforms the existing IRSTD methods and achieves state-of-the-art (SOTA) detection performance. The code will be released on https://github.com/Wangtao-Bao/HAFNet**

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***Index Terms*—Dual-branch semantic perception module (DSPM), hierarchical attention fusion network (HAFNet), infrared small target detection (IRSTD), skip connections.**

I. INTRODUCTION

INFRARED small target detection (IRSTD) refers to the task of accurately identifying and localizing small-sized, weak-contrast targets within infrared images. This task plays a critical role in applications including early warning systems, maritime search, and rescue [1]. Compared to general object detection tasks in visible-spectrum imagery, IRSTD presents unique challenges, including low resolution, the extremely small target-to-background ratio, and background interference, such as sea surfaces or cloud structures [4]. These characteristics make IRSTD a highly underdetermined and noise-sensitive problem, requiring detection methods that are specifically designed for enhanced robustness, fine-grained feature extraction, and effective background suppression.

Traditional IRSTD methods largely relied on handcrafted features, such as grayscale differences between target and background, local contrast enhancement, and texture extraction. Well-known techniques include filter-based methods [6], the human visual system (HVS)-based methods [9], and low-rank-based methods [12]. However, these methods often exhibit high false alarm rates and missed detections, particularly in complex backgrounds or weak signal conditions. In addition, these approaches fail to adapt to dynamic and diverse environmental conditions, resulting in poor stability in real-world scenarios.

The rapid development of deep learning techniques has provided promising solutions to the IRSTD task by automatically learning feature representations from data samples, thereby eliminating the requirement for handcrafted features. However, these methods face challenges due to the relatively smaller size of infrared targets compared to general objects, which complicates accurate detection. In addition, conventional detection models based on bounding box predictions [15] are prone to target loss when applied to small targets, further hindering detection. U-Net [17], a widely recognized architecture for image segmentation, has gained significant attention for IRSTD owing to its effectiveness in capturing spatial features through its encoder–decoder structure with skip connections. Recent modifications to U-Nets, such as the incorporation of asymmetric modules [18] and optimization of skip connections [19], have led to substantial advancements in IRSTD performance, particularly in complex infrared scenes.

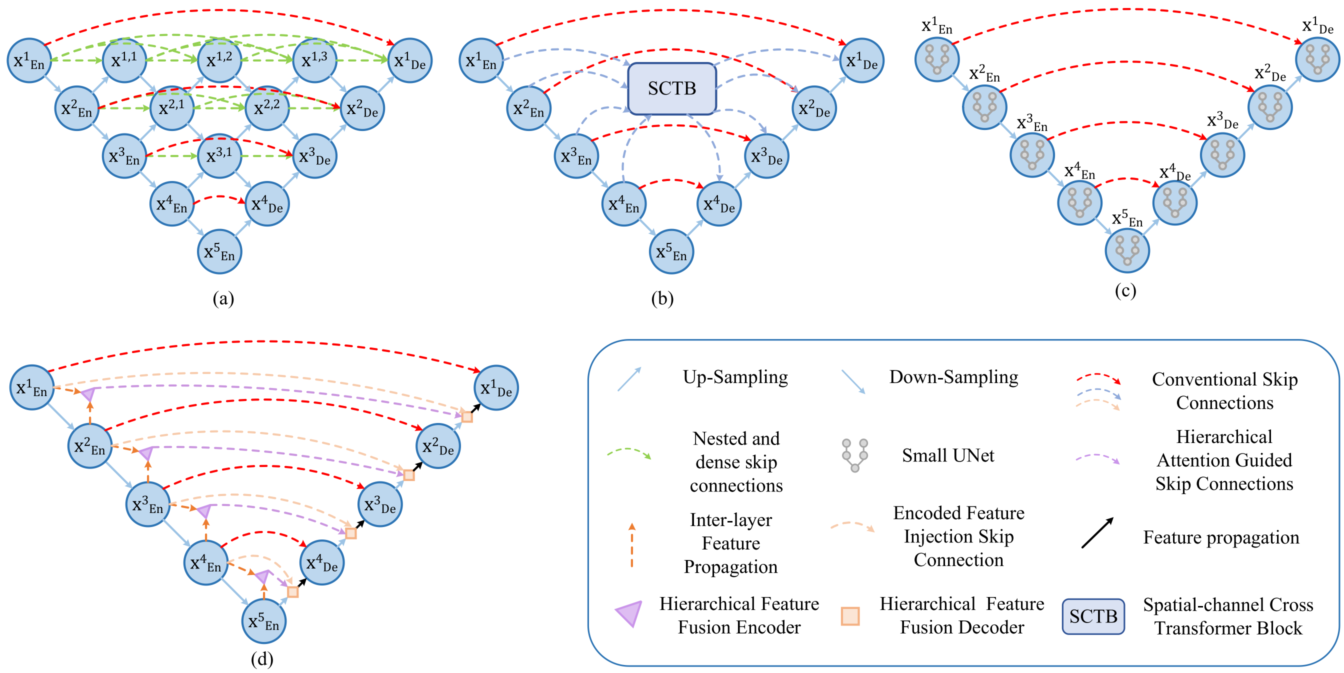

Fig. 1. IRSTD methods based on various U-Net models. (a) DNA-Net 0. (b) SCTransNet 0. (c) UIUNet 0. (d) HAFNet (Ours).

Despite these advancements, the standard U-Net architecture still faces challenges in detecting small targets amidst intricate and cluttered backgrounds.

To handle this problem, several U-Net variants have been proposed, each introducing specific improvements. DNA-Net [22], shown in Fig. 1(a), introduces nested and dense skip connections by stacking multiple U-Net subnetworks to enable more effective feature fusion at different levels. This design improves small target detection as it facilitates the aggregation of features across different levels, thereby enhancing the representation of both fine-grained and broader contextual information. SCTransNet [23], shown in Fig. 1(b), incorporates spatial–channel cross transformer blocks (SCTBs) into the U-Net framework to enhance target-background discrimination by capturing global context and long-range dependencies. The SCTBs help reduce false positives, especially in noisy environments, by enabling the network to focus on relevant features and suppress background interference. UIUNet [24], shown in Fig. 1(c), introduces a smaller U-Net embedded within a larger U-Net, facilitating multiscale and multilevel feature aggregation, which significantly improves detection of small targets at various scales. While these improvements have advanced the performance of U-Net in IRSTD, challenges remain. One key issue is the reliance on classification backbones, such as Resnet-20 in ACM [18] and ALC-Net [25]; Resnet-10/18/34 in MTU-Net [19], Resnet-18 in DMFN-et [26], and Resnet-18/34 in DNA-Net [22]; and AGPC-Net [27], which are primarily designed for visible spectrum data and often exhibit poor generalization to infrared imagery due to substantial spectral disparities. In addition, the traditional skip connections in U-Net, which perform feature fusion at identical scale, frequently lead to suboptimal integration owing to semantic discrepancies between high- and low-level features [28].

However, two critical limitations persist in existing approaches. First, the feature extraction modules in many methods rely on classification-oriented backbones, which are originally designed for visible-spectrum images. Due to the significant spectral and distributional differences between visible and infrared imagery, these backbones often fail to generalize effectively in IRSTD scenarios, resulting in suboptimal feature representations and reduced robustness in complex infrared environments. Second, conventional skip connections (CSCs) in U-Net directly propagate features between encoder and decoder layers with matching spatial resolution. This strategy overlooks the inherent semantic gap between low-level features, which are rich in spatial detail but susceptible to noise, and high-level features, which are semantically abstract but lack fine-grained structural information. As a result, the fusion of such heterogeneous features is often ineffective, thereby limiting detection accuracy, especially for small targets in cluttered or highly textured backgrounds.

To address the limitations of previous U-Net-based methods, the hierarchical attention fusion network (HAFNet) is proposed, shown in Fig. 1(d), which is a novel extension of U-Net that incorporates hierarchical feature fusion and attention-guided skip connections, inspired by successful U-Net variants [29], [30], [31], [32] and attention mechanisms [33], [34] in image processing. Specifically, HAFNet consists of three core modules: the dual-branch semantic perception module (DSPM), the hierarchical feature fusion encoder (HFFE), and the hierarchical feature fusion decoder (HFFD). Unlike traditional residual blocks [18], [23], [26], DSPM integrates standard and dilated convolutions to expand the receptive field and capture multiscale contextual information, enabling the network to extract both local and global features effectively. Furthermore, a novel mechanism with a hierarchical attention-guided skip connection (HSC) and encoded feature injection skip connection (ESC) is designed, which transfers encoded features from the encoder to the decoder using hierarchical feature fusion. These connections allow the decoder to flexibly integrate feature maps of different scales and semantic levels, mitigating semantic discrepancies and redundancy between high- and low-level features, thereby enhancing the recovery of small targets. Extensive evaluation of HAFNet on multiple datasets and backbone networks demonstrates significant improvements in detection accuracy and a reduction in false alarm rates, outperforming state-of-the-art (SOTA) methods.

In summary, the main contributions of this work are as follows.

1) We propose HAFNet, which improves the original U-Net architecture from two aspects: feature extraction backbone and skip connection, aiming to enhance the detection ability and generalization performance.

2) We construct DSPM to enhance the feature extraction capability of the network. The dual-branch feature extraction of standard and dilated convolutions and the feature enhancement of attention significantly improve the network's semantic perception ability of infrared small targets.

3) We design HFFE and HFFD to introduce new skip connections and combine them with traditional skip connections to achieve flexible feature fusion in the decoder, alleviating the limitation of the original U-Net architecture that only fuses feature maps of the same scale.

4) Extensive experiments were conducted on multiple public datasets and backbone networks of different depths, and the experimental results demonstrate that the proposed method is superior to the SOTA methods, fully verifying the effectiveness of HAFNet.

The rest of this article is organized as follows. Section II briefly reviews related work. Section III details the network architecture and the design of DSPM, HFFE, and HFFD. Section IV presents experimental settings and results, and Section V concludes this article.

II. Related Work

A. IRSTD Methods

In recent years, numerous methods have been proposed for IRSTD, including traditional methods and deep learning methods. Traditional methods typically leverage the physical differences between the target and the background to separate the target. These methods can be roughly classified into three categories: filter-based methods, HVS-based methods, and low-rank-based methods. Filter-based methods [6], [7], [8] enhance target-background contrast through various filtering operations, thereby facilitating target extraction. HVS-based methods [9], [10], [11] detect small targets by simulating the perceptual characteristics of the HVS. Low-rank-based methods [12], [13], [14] model the background as a low-rank matrix and characterize the target as a sparse, high-rank component, enabling effective separation of the background from the target through low-rank decomposition. Although traditional methods have undergone significant development and provide valuable insights for small target detection, they still face persistent challenges, particularly in the context of complex backgrounds and noise interference. These challenges are especially evident in dynamic scenes, where their robustness remains inadequate, thereby limiting their practical applicability.

With the rapid development of deep learning technology, IRSTD has overcome the performance limitations of traditional methods, achieving notable advancements. ACM [18] proposed a pioneering IRSTD framework based on image segmentation and released the NUAA-SIRST dataset, establishing a standardized benchmark for subsequent research. ISNet [35] introduced edge information and improved the detection accuracy through an infrared shape reconstruction mechanism. FC3-Net [36] investigated feature compensation and cross layer information correlation to boost IRSTD performance. MTUNet [19] designed a multilevel TransU-Net architecture tailored for infrared ship detection, improving detection accuracy through background modeling.

In addition, Dim2Clear [37] extracted discriminative target features of the target via interactive guidance between high- and low-level feature maps; GCI-Net [38] leveraged a Gaussian-based approach to design a structural information extraction branch. HAMANet [30] proposed a multilevel and multiscale context fusion mechanism to enhance both robustness and real-time performance. CFD-Net [39] reevaluated the IRSTD feature learning process from a pixel-level motion modeling perspective. Recently, researchers have also explored adapting general segmentation models to IRSTD, such as IRSAM [40] and SimIRSTD [41], and improving inference efficiency through model pruning techniques, exemplified by IRPruneDet [42].

B. U-Net Variants in Image Processing

U-Net is a well-established architecture based on a fully convolutional neural network, initially proposed for medical image segmentation. Its symmetric encoder–decoder structure facilitates efficient feature extraction and multiscale information fusion, making it highly versatile for various segmentation tasks [43]. Numerous U-Net variants have been developed, primarily focusing on two key improvements: enhancing feature extraction capabilities and optimizing skip connection mechanisms.

Regarding feature extraction, several U-Net variants have been proposed to bolster the network's capacity to capture complex patterns. ResUNet [44] integrated residual connections from ResNet [45] into both the encoder and decoder, enabling deeper network architectures while alleviating vanishing gradient issues and training difficulties. MultiResUNet [46] embedded inception-like modules into the U-Net architecture to enhance the multiscale feature extraction capabilities. DenseUNet [47] adopted the densely connected structure of DenseNet [48], promoting enhanced feature reuse and extraction efficiency. Furthermore, TransUNet [49] combined the global information capture abilities of transformers with the local feature extraction strength of CNNs, leading to

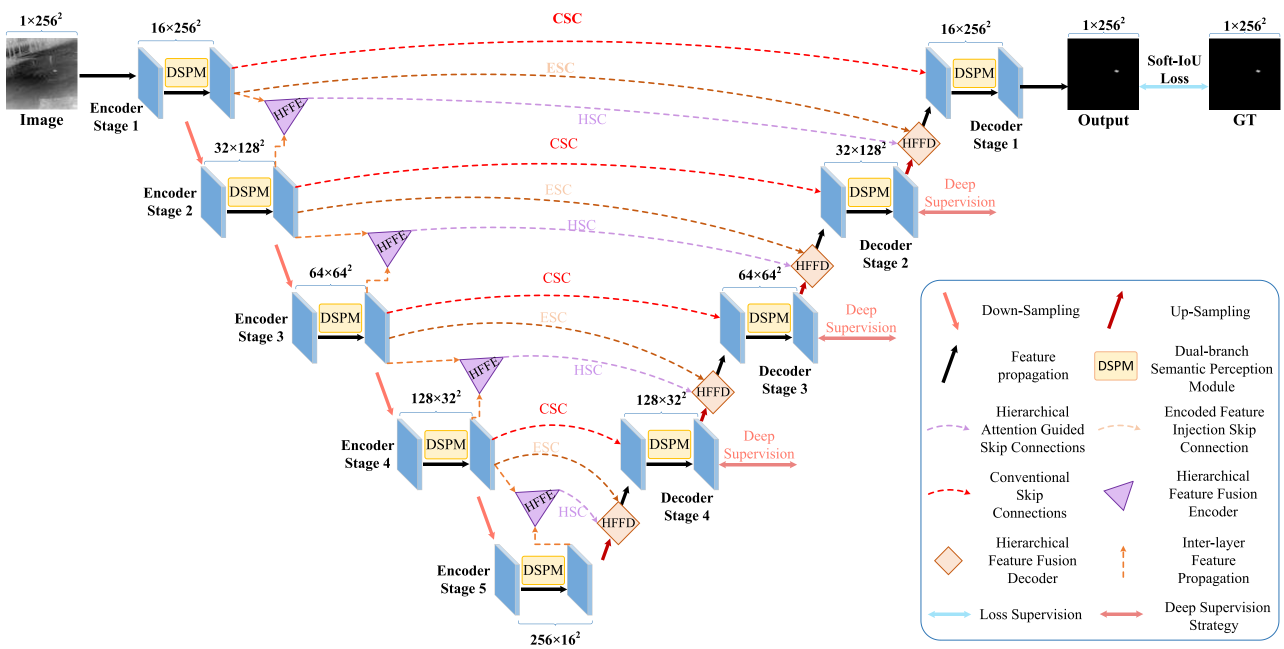

Fig. 2. Detailed schematic of the proposed HAFNet. HAFNet contains three specially designed modules: DSPM, HFFE, and HFFD. HAFNet also has three kinds of skip connections: CSC, ESC, and HSC. For simplicity, we only use arrows to represent DSPM.

significant improvements in segmentation performance. However, since the traditional transformer block failed to fully exert its advantages, Swin-Unet [50] replaced the standard transformer with a Swin transformer [51] and designed a symmetrical Swin-like decoder, enabling the network to more meticulously restore fine-grained details in multiple organ segmentation.

In the context of optimizing skip connections, several approaches have been proposed to refine feature transfer between the encoder and decoder. The original U-Net employs direct skip connections between corresponding encoder and decoder layers. Attention U-Net [31] introduced an attention mechanism (attention gate) to selectively transmit essential features from the encoder to the decoder, focusing on relevant information while filtering out noise. CU-Net [52] improved brain tumor segmentation by utilizing a cascading structure that transfers high-resolution features from shallow layers to deeper ones, refining segmentation accuracy. U-Net++ [53] replaced traditional skip connections with dense connections, which facilitate multipath feature aggregation and enhance semantic consistency. U-Net 3+ [54] introduced full-scale skip connections, enabling the transfer of encoder features of the same or smaller scale to the decoder. In addition, it incorporates dense connections between decoders to facilitate feature fusion, thereby enhancing full-scale information integration. Finally, AHF-U-Net [32] incorporated hierarchical attention within skip connections, ensuring precise spatial information transfer from encoder to decoder, further enhancing segmentation accuracy.

III. Method

A. Network Architecture

The objective of the proposed method is to detect infrared small targets, which are typically characterized by their tiny spatial scale, weak contrast, and presence within complex backgrounds, such as sea surfaces, clouds, or terrain. These challenges necessitate a detection framework capable of enhancing subtle target features while effectively suppressing background noise.

To address this, we propose HAFNet, designed for IRSTD. As illustrated in Fig. 2, this network adopts a hierarchical encoder–decoder architecture composed of three core modules: the DSPM, HFFE, and HFFD. The DSPM serves as the backbone for feature extraction, leveraging both standard and dilated convolutions to balance local detail preservation and global context awareness, thereby enabling precise and robust feature representation. Unlike traditional U-Net architectures, HAFNet introduces an additional hierarchical encoding branch in the encoder stage. The HFFE module fuses adjacent encoder features to form hierarchical representations, which are transmitted to the HFFD module through HSC and ESC. The HFFD module then captures multiscale contextual information and produces hierarchical decoding features, which are further refined through CSCs before being fed into the decoder for target reconstruction. Finally, the fused decoder features are delivered to the detection head to generate the final predictions.

Specifically, given an infrared input image $I \in \mathbb{R}^{H\times W\times C}$, the encoding stage employs five groups of DSPM and max-pooling layers to extract encoded features $X_{en}^{i} (i = 1, 2, \ldots, 5)$,

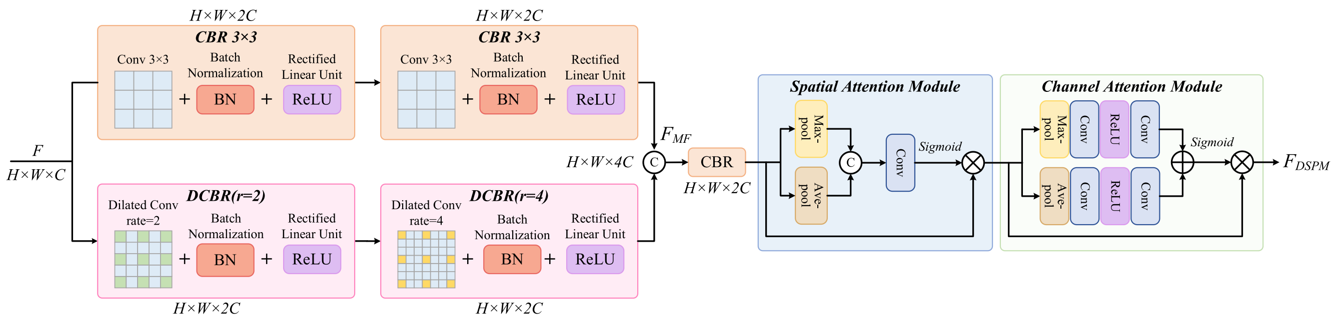

Fig. 3. Detailed schematic of the proposed DSPM.

where the resolution reduces as $(H/2^{i-1}) \times (W/2^{i-1})$ with increasing depth. The number of feature channels for each layer is defined as $C = [16, 32, 64, 128, 256]$. The hierarchical encoding features X^i_{HFFE} are generated by the HFFE module, mathematically expressed as

$$X^i_{\text{en}} = \text{DSPM}\left(\mathcal{P}_{\max}\left(X^{i-1}_{\text{en}}\right)\right), \quad i \in [1, 2, 3, 4, 5] \quad (1)$$

$$X^i_{\text{HFFE}} = \text{HFFE}\left(X^i_{\text{en}}, \mathcal{B}i\left(X^{i+1}_{\text{en}}\right)\right), \quad i \in [1, 2, 3, 4] \quad (2)$$

where $\mathcal{P}_{\max}$ represents a max-pooling layer with a kernel size and stride of 2, and $\mathcal{B}i$ denotes bilinear interpolation for $2\times$ upsampling. When $i = 1, X^0_{\text{en}}$ corresponds to the infrared input image I.

In the decoding stage, the hierarchical decoding features X^j_{HFFD} are obtained by fusing upsampling decoder features, hierarchical encoding features, and encoder features through the HFFD module. Simultaneously, the decoded features X^j_{de} are refined by DSPM, incorporating CSC to facilitate detailed reconstruction. These processes are expressed as

$$X^j_{\text{HFFD}} = \text{HFFD}\left(X^j_{\text{en}}, X^j_{\text{HFFE}}, \mathcal{B}\text{i}\left(X^{j+1}_{\text{de}}\right)\right), \quad j \in [4, 3, 2, 1] \quad (3)$$

$$X^j_{\text{de}} = \text{DSPM}\left[X^j_{\text{en}}, X^j_{\text{HFFD}}\right], \quad j \in [4, 3, 2, 1]. \quad (4)$$

When $j = 4, X^5_{\text{de}}$ corresponds to the encoding feature of the fifth layer, where $X^5_{\text{de}} = X^5_{\text{en}}$. The notation $[\cdot]$ denotes the concatenation operation along the channel dimension.

To further enhance the network's feature learning capability at different scales, a deep supervision (DS) strategy is introduced for optimization. Each decoded feature X^j_{de} is input into a detection head consisting of a 1×1 convolutional layer and a sigmoid activation function to generate the corresponding segmentation map. Supervision is applied to these maps using the Soft-IoU loss [55] to effectively optimize the segmentation results and minimize false positives.

B. Dual-Branch Semantic Perception Module

Traditional convolutional operations are inherently constrained by their fixed receptive field, limiting their capacity to capture long-range dependencies. Standard 3×3 or 5×5 convolutions are primarily effective for extracting local spatial information, yet they struggle to distinguish small infrared targets from complex backgrounds effectively. Noise interference and variations in target contrast further exacerbate this issue, posing significant challenges to robust small target detection. In addition, infrared small targets exhibit diverse scale distributions across different scenes, rendering single-scale feature extraction methods inadequate for capturing multiscale target variations.

To address these challenges, we propose the DSPM as the foundational feature extraction module within HAFNet. DSPM enhances the receptive field and feature representation by integrating multiscale feature extraction with spatial and channel attention mechanisms, as illustrated in Fig. 3. It consists of two core components: dual-branch feature extraction and dual attention modules. Given an input feature map $F \in \mathbb{R}^{H\times W\times C}$, DSPM employs a dual-branch structure to extract multiscale features. The first branch applies a standard convolution operation to preserve local spatial consistency and extract fundamental low-level features. The second branch incorporates dilated convolutions with multiple dilation rates to expand the receptive field and capture contextual information across different scales. This dual-branch approach enables DSPM to grasp contexts of different scales, greatly enriching the feature representation and ensuring that the network establishes a balance between local details and global context. The feature extraction process is formulated as

$$F_{\text{MF}} = \text{CBR}_1\left[\text{CBR}_3\left(\text{CBR}_3\left(F\right)\right), \text{DCBR}_{r=2}\left(\text{DCBR}_{r=4}\left(F\right)\right)\right] \quad (5)$$

where F_{MF} represents the extracted multiscale features, CBR_1 represents a 1×1 convolution followed by batch normalization and a ReLU activation function, and CBR_3 represents a standard 3×3 convolutional layer. $\text{DCBR}_{r=2}$ and $\text{DCBR}_{r=4}$ correspond to the dilated convolution with a dilation rate of 2 and 4, respectively.

After multiscale feature extraction, DSPM applies spatial and channel attention mechanisms to refine the fused features, ensuring that the network focuses on relevant target regions while suppressing background noise. The process is described as

$$F_{\text{DSPM}} = C_{\text{att}}\left(S_{\text{att}}\left(F_{\text{MF}}\right)\right) \quad (6)$$

where F_{DSPM} denotes the output of DSPM, and S_{att} and C_{att} represent the spatial attention module (SAM) [33] and channel attention module (CAM) [33], respectively. More

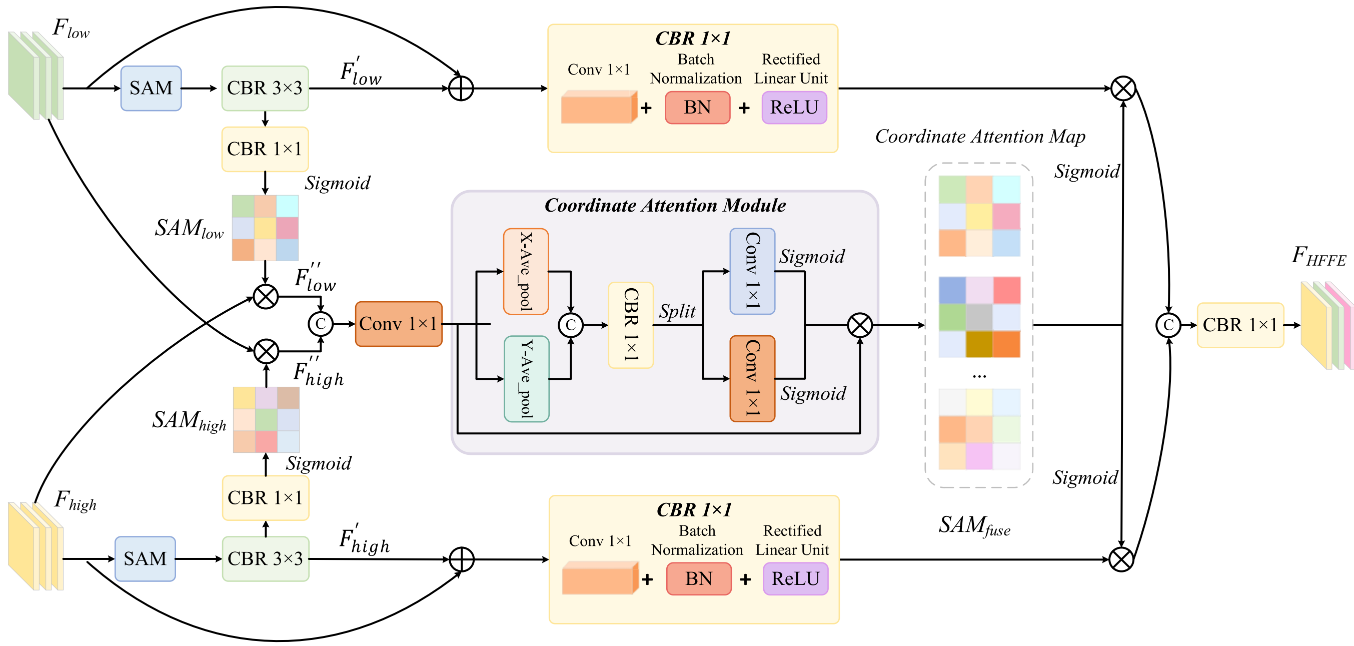

Fig. 4. Detailed schematic of the proposed HFFE.

specifically, the spatial and channel attention operations can be implemented as follows:

$$S_{\text{att}}(F_{\text{in}}) = \sigma\left(\text{Conv}_{7\times 7}\left[\mathcal{P}_{\max}(F_{\text{in}})\,\mathcal{P}_{\text{avg}}(F_{\text{in}})\right]\right) \otimes F_{\text{in}} \quad (7)$$

$$C_{\text{att}}(F_{\text{in}}) = \sigma\begin{pmatrix}\text{MLP}\left(\mathcal{P}_{\max}(F_{\text{in}})\right) \\ +\text{MLP}\left(\mathcal{P}_{\text{avg}}(F_{\text{in}})\right)\end{pmatrix} \otimes F_{\text{in}} \quad (8)$$

where F_{in} represents the given input features, σ represents the sigmoid function, $\text{Conv}_{7\times 7}$ represents the convolution operation with a kernel size of 7×7, $\mathcal{P}_{\text{avg}}$ represents the average-pooling operation with a stride of 2, and $\otimes$ represents the element-wise multiplication. The shared network consists of an MLP with one hidden layer.

By leveraging a dual-branch feature extraction structure and attention-guided refinement, DSPM enables robust multiscale feature representation, facilitating the effective distinction of small infrared targets from complex backgrounds. The combination of dilated convolutions for context enhancement and attention mechanisms for target-focused feature refinement significantly improves detection performance in challenging infrared scenes.

C. Hierarchical Feature Fusion Encoder

Although DSPM enhances feature extraction by expanding the receptive field and capturing contextual information, it lacks pertinence in distinguishing pseudotargets and suppressing background interference, leading to false detections. Furthermore, traditional U-Net-based encoders extract multilevel features with distinct statistical properties [56]. Low-level features preserve fine-grained textures and edges but are susceptible to noise, whereas high-level features provide a comprehensive semantic representation but suffer from a loss of fine structural details. The primary challenge lies in effectively leveraging multilevel feature representations to improve target localization and background suppression.

To address the above challenges, an HFFE module is proposed, integrating adjacent encoder features through a cross attention mechanism for cross layer fusion. Unlike conventional U-Net architectures that rely solely on direct feature concatenation, HFFE enhances feature representation by leveraging both SAM and coordinate attention (CoordAtt) to selectively emphasize target-relevant features while suppressing noise.

As illustrated in Fig. 4, HFFE takes encoding features from the previous level F_{low} and the current level F_{high} as inputs.

Given the resolution discrepancy between these feature maps, F_{high} undergoes bilinear upsampling to match the spatial dimensions of F_{low}. Each feature map is then independently processed using SAMs to highlight target-relevant regions, producing refined feature representations

$$F'_{\text{low}} = \text{CBR}_3\left(S_{\text{att}}(F_{\text{low}})\right) \quad (9)$$

$$F'_{\text{high}} = \text{CBR}_3\left(S_{\text{att}}\left(\mathcal{B}i\left(F_{\text{high}}\right)\right)\right) \quad (10)$$

where S_{att} represents the spatial attention operation.

To ensure effective cross layer feature interaction, HFFE generates spatial weight matrices (SWMs), which adaptively recalibrate feature importance across different scales. These matrices are computed using channel-wise recalibration and are formulated as follows:

$$\text{SWM}_{\text{low}} = \sigma\left(\text{CBR}_1\left(F'_{\text{low}}\right)\right) \quad (11)$$

$$\text{SWM}_{\text{high}} = \sigma\left(\text{CBR}_1\left(F'_{\text{high}}\right)\right) \quad (12)$$

where SWM represents the generated spatial weight matrix.

The recalibrated feature maps are then obtained through element-wise multiplication with their respective SWMs

$$F''_{\text{low}} = F_{\text{low}} \otimes \text{SWM}_{\text{high}} \quad (13)$$

$$F''_{\text{high}} = F_{\text{high}} \otimes \text{SWM}_{\text{low}}. \quad (14)$$

To further enhance the synergy between different features, the coordinate attention (CoordAtt) [34] is incorporated, which encodes spatial dependencies along both the horizontal and vertical directions through a global pooling operation. Unlike conventional channel attention mechanisms, CoordAtt retains positional information crucial for small target detection. The fused coordinate attention weights are generated as

$$\mathrm{SWM}_{\mathrm{fuse}} = \sigma\left(\mathrm{CA}\left(\mathrm{Conv}_{1\times 1}\left[F'_{\mathrm{low}}, F'_{\mathrm{high}}\right]\right)\right). \tag{15}$$

The final hierarchical encoding feature F_{HFFE} is then computed as

$$F_{\mathrm{HFFE}} = \mathrm{Conv}_{1\times 1}\begin{bmatrix} \mathrm{SWM}_{\mathrm{fuse}} \otimes \left(\mathrm{CBR}_1\left(F'_{\mathrm{low}} + F_{\mathrm{low}}\right)\right) \\ \mathrm{SWM}_{\mathrm{fuse}} \otimes \left(\mathrm{CBR}_1\left(F'_{\mathrm{high}} + F_{\mathrm{high}}\right)\right) \end{bmatrix}. \tag{16}$$

By integrating HFFE into the network, the model effectively leverages the statistical differences among multilevel encoder features, ensuring robust feature aggregation and noise suppression. Compared to traditional skip connections, HFFE refines feature interactions at multiple scales, reducing false positives caused by DSPM while enhancing the discriminative capacity of hierarchical encoding features. The refined hierarchical representations generated by HFFE are subsequently fed into the decoder stage through HSC, where they further enhance target localization and improve detection accuracy in complex infrared scenes.

D. Hierarchical Feature Fusion Decoder

Skip connections play a crucial role in U-shaped networks, facilitating feature propagation between the encoder and decoder while mitigating information loss during upsampling. However, traditional U-Net architectures rely on direct skip connections at the same scale, which often struggle to capture global semantic context and fine-grained local details concurrently. As a result, the decoder encounters challenges in accurately reconstructing target structures, particularly in complex infrared scenes where small targets may be obscured by background noise. To enhance the decoder's ability to reconstruct targets with high precision, we introduce two additional skip connections alongside conventional ones. Inspired by [53], the HFFD is proposed to maximize the utility of multilevel features. By integrating multisource information through a structured fusion process, HFFD ensures efficient information transmission, significantly improving the decoder's capacity for small target restoration.

As depicted in Fig. 5, the HFFD module receives three input features and fuses them, including:

1) encoder-extracted features (F_{en}), which provide fine-grained local details necessary for precise target boundary reconstruction;
2) hierarchical encoding features (F_{HFFE}), which aggregate multiscale encoder representations, enhancing global semantic understanding; and
3) upsampled decoder features (F_{de}), which are derived from the previous decoder layer and contribute spatial context for the current layer decoding to guide the reconstruction process.

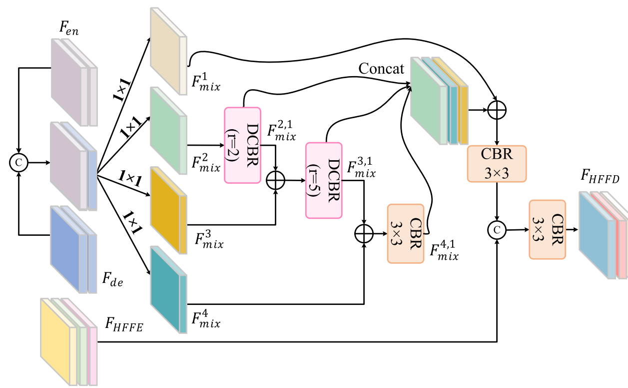

Fig. 5. Detailed schematic of the proposed HFFD.

To effectively merge these heterogeneous feature sources, the decoder first concatenates F_{en} and F_{de} along the channel dimension to form a fused feature F_{mix}, which undergoes feature decomposition (FD) through four convolutional layers with a kernel size of 1×1 to generate multiple feature maps, expressed as

$$F^{i}_{\mathrm{mix}} = \mathrm{Conv}_{1\times 1}\left[F_{\mathrm{en}}, F_{\mathrm{de}}\right], \quad i = 1, 2, 3, 4. \tag{17}$$

Subsequently, multiscale feature enhancement is performed using convolutional layers with different dilation rates, allowing the decoder to capture both localized structures and long-range dependencies. The multiscale features are progressively refined through element-wise summation operations across different dilation levels

$$\begin{cases} F^{2,1}_{\mathrm{mix}} = \mathrm{DCBR}_{r=5}\left(F^{2}_{\mathrm{mix}}\right) \\ F^{3,1}_{\mathrm{mix}} = \mathrm{DCBR}_{r=2}\left(F^{2,1}_{\mathrm{mix}} + F^{3}_{\mathrm{mix}}\right) \\ F^{4,1}_{\mathrm{mix}} = \mathrm{CBR}_3\left(F^{3,1}_{\mathrm{mix}} + F^{4}_{\mathrm{mix}}\right). \end{cases} \tag{18}$$

The extracted multiscale features are then concatenated along the channel dimension and further refined via a 3×3 convolutional layer, producing a cascaded representation. To preserve critical information from the initial feature interactions, the cascaded features are combined with F^{1}_{mix} through element-wise summation. Finally, a 3×3 convolutional layer is employed to integrate the enhanced feature representations with the hierarchical encoding features F_{HFFE} to generate the final hierarchical decoding features F_{HFFD}, expressed as

$$F_{\mathrm{HFFD}} = \mathrm{CBR}_3\left[\mathrm{CBR}_3\left(F^{1}_{\mathrm{mix}} + \left[F^{2,1}_{\mathrm{mix}}, F^{3,1}_{\mathrm{mix}}, F^{4,1}_{\mathrm{mix}}\right]\right), F_{\mathrm{HFFE}}\right]. \tag{19}$$

By leveraging the HFFD module, the network comprehensively utilizes multilevel encoder features, enhances cross layer interactions, and effectively suppresses background noise. Compared to conventional decoder designs that rely on direct skip connections, HFFD synergistically fuses complementary feature representations, facilitating high-fidelity infrared target restoration even in cluttered backgrounds.

IV. Experiments

A. Datasets and Data Preprocessing

Datasets: To comprehensively evaluate the effectiveness and superiority of the proposed HAFNet, three widely

TABLE I
CUSTOM HYPERPARAMETERS FOR FOUR DATASETS

Dataset	Epoch	Lr	Batch	Resolution	Ratio	Dataset	Epoch	Lr	Batch	Resolution	Ratio
NUAA-SIRST [18]	1000	0.001	4	320×320	1:1	IRSTD-1K [35]	1000	0.001	4	512×512	4:1
NUDT-SIRST [22]	1000	0.001	8	256×256	1:1	NoisySIRST [21]	1000	0.001	16	256×256	4:1

recognized benchmark datasets are employed: NUAA-SIRST [18], IRSTD-1K [35], and NUDT-SIRST [22]. These datasets contain 427, 1001, and 1327 infrared images, respectively, with corresponding annotations. They encompass a diverse range of infrared scenes, ensuring a comprehensive and representative evaluation environment. The variability in target scale, contrast, and background complexity across these datasets provides a robust basis for validating HAFNet's performance across different scenarios. In addition, to further evaluate the detection performance of HAFNet in a noisy environment, we conducted comparative experiments with other deep learning-based methods. The experiment is based on the NoisySIRST dataset [21], which simulates the actual situation of infrared image quality degradation under severe noise interference by injecting Gaussian white noise of different intensities (σ_n = 10, 20, and 30) into the NUAA-SIRST dataset. The training set and test set division of all methods follows the setting ratio (4:1) of the original paper. We train independent models on these six datasets to ensure the reliability and accuracy of the evaluation.

Data Preprocessing: During the construction of training data, all images are converted to grayscale and normalized based on the mean and standard deviation of the training set to standardize the data distribution and accelerate model convergence. To enhance robustness in complex scenarios, a mask-based random cropping strategy is employed to select positive patches (i.e., image patches containing targets) with a probability of 0.5, ensuring the inclusion of target regions to improve the model's ability to capture small targets. In addition, to further boost generalization, we apply standard data augmentation techniques, such as random rotation and horizontal flipping. For images with insufficient spatial dimensions, zero-padding is applied to the shorter sides to ensure consistent input dimensions for the network.

B. Implementation Details

During the training phase, we select AdamW [58] as the optimizer with an initial learning rate of 0.001 and a weight decay of 10^{-2}. A cosine annealing strategy is applied to gradually reduce the learning rate to 1×10^{-5}, ensuring stable convergence. The model weights and biases are initialized using the Kaiming initialization method [59]. Given the substantial variations in data distribution and resolution across different datasets, we finetune hyperparameters for each dataset, as summarized in Table I. In the inference phase, model predictions are resized to match the original image resolution, and performance evaluation is conducted using multiple quantitative metrics. Following prior works [18], [22], [25], the segmentation threshold for saliency maps is set to 0.5.

C. Evaluation Metrics

To provide a comprehensive evaluation of detection performance, we employ five widely used metrics: intersection over union (IoU), normalized IoU (nIoU) [18], detection probability (P_d), false alarm rate (F_a), and F1 Score ($F1$) for the experiment. They are defined as follows:

$$\text{IoU} = \frac{TP}{T + P - TP} \tag{20}$$

$$\text{nIoU} = \frac{1}{N}\sum_{i}^{N}\frac{TP(i)}{T(i) + P(i) - TP(i)} \tag{21}$$

$$P_d = \frac{TP}{TP + \text{FN}} \tag{22}$$

$$F_a = \frac{\text{FP}}{\text{FP} + TN} \tag{23}$$

$$F1 = \frac{2TP}{2TP + \text{FP} + \text{FN}} \tag{24}$$

where N is the total number of samples, an T and P represent the number of true values and predicted positive pixels, respectively. TP, FP, and FN denote the number of true positive, false positive, and false negative pixels, respectively. Given that IRSTD is often affected by high background clutter, we additionally employ the receiver operating characteristic (ROC) curve to analyze the trade-off between the true positive rate (TPR) and the false positive rate (FPR), providing a more comprehensive evaluation of model performance.

D. Comparison With SOTA Methods

We compare the proposed HAFNet with 13 existing IRSTD methods, comprising six traditional methods (Top-Hat [8], Max-Median [7], WSLCM [60], TLLCM [61], IPI [62], and NOLC [63]), and seven deep learning-based methods (ACM [18], RDIAN [64], DNANet [22], UIUNet [24], RPCANet [65], MSHNet [66], and SCTransNet [23]). To ensure a fair comparison, we retrain all deep learning models using the same training benchmark as that used for HAFNet. The training process strictly follows the settings described in the original papers, and hyperparameters are kept consistent across all methods to eliminate potential biases.

E. Quantitative Results

Table II presents a comprehensive comparison of HAFNet with existing IRSTD methods on three benchmark datasets: NUAA-SIRST, IRSTD-1K, and NUDT-SIRST. The experimental results demonstrate that HAFNet consistently achieves the best performance across all evaluation metrics, fully verifying its effectiveness in detecting infrared small targets. The superior performance can be attributed to the enhanced

TABLE II

COMPARISON WITH SOTA METHODS ON THREE DATASETS. IN IoU (%)↑, nIoU (%)↑, P_d(%)↑, F_a(10^{-6})↓, AND $F1$(%)↑

Methods	NUAA-SIRST					IRSTD-1K					NUDT-SIRST				
	IoU	nIoU	P_d	F_a	$F1$	IoU	nIoU	P_d	F_a	$F1$	IoU	nIoU	P_d	F_a	$F1$
Traditional Methods															
Top-Hat [8]	7.14	18.27	79.84	1012	14.63	10.06	7.44	75.11	1432	16.02	20.72	28.98	78.41	166.7	33.52
Max-Median [7]	4.17	12.31	69.20	55.53	10.67	7.00	3.05	65.21	59.73	8.15	4.20	3.67	58.41	36.89	7.64
WSLCM [60]	1.16	6.84	77.95	5446	4.81	3.45	0.68	72.44	6619	2.13	2.28	3.87	56.82	1309	5.99
TLLCM [61]	1.03	4.10	79.09	5899	5.00	3.31	0.78	77.39	6738	2.19	2.18	4.32	62.01	1608	7.23
IPI [62]	25.67	50.17	84.63	16.67	43.65	27.92	20.46	81.37	16.18	35.68	17.76	15.42	74.49	41.23	26.94
NOLC [63]	17.70	26.22	89.69	16.41	30.08	20.46	28.28	82.82	49.64	33.97	18.61	26.65	85.61	18.19	31.38
Deep Learning Methods															
ACM [18]	65.28	65.67	90.49	47.13	78.99	58.84	58.23	91.92	27.59	74.09	60.63	60.94	91.75	20.06	75.49
RDIAN [64]	70.66	74.04	93.92	43.08	82.80	63.37	62.63	88.55	30.18	77.59	77.77	79.46	95.66	26.36	87.49
DNANet [22]	76.34	79.19	95.82	15.23	86.12	66.50	66.13	91.58	18.31	79.89	91.96	92.88	98.94	9.28	95.81
UIUNet [24]	77.17	78.86	95.44	14.34	87.11	65.34	65.55	90.91	15.01	79.02	90.43	89.87	98.94	7.72	94.97
RPCANet [65]	53.93	61.01	95.44	149.2	70.07	62.09	61.48	89.35	48.61	76.61	87.79	89.80	95.98	32.03	93.50
MSHNet [66]	73.88	72.56	95.82	19.25	84.98	67.17	60.38	92.52	12.60	80.35	76.95	79.72	95.77	20.52	86.97
SCTransNet [23]	75.46	78.06	95.82	18.80	86.01	66.54	66.36	93.27	13.68	79.91	93.01	93.50	98.73	4.80	96.38
HAFNet(Ours)	**79.19**	**81.00**	**97.72**	**14.06**	**88.39**	**67.95**	**69.23**	**94.61**	**10.57**	**80.91**	**96.28**	**96.11**	**99.26**	**1.79**	**98.10**

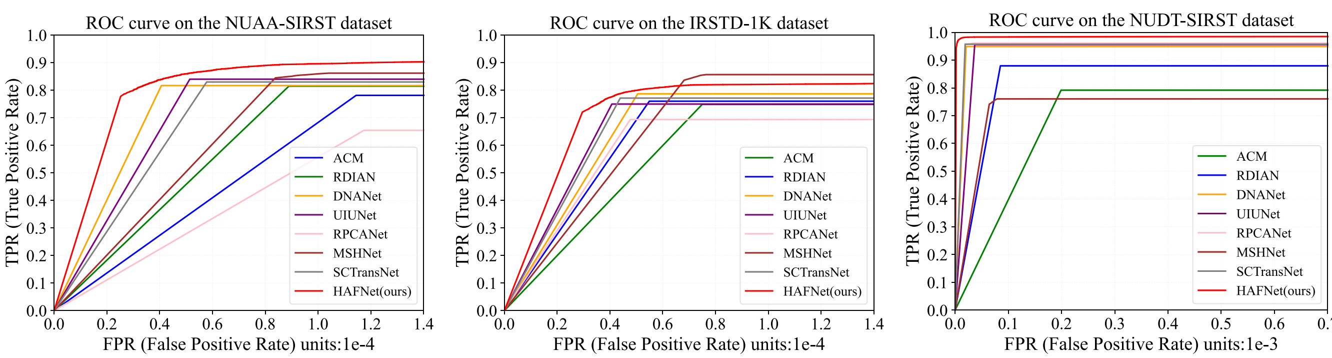

Fig. 6. ROC curves of different IRSTD methods on three public datasets.

feature extraction capability and the optimized skip connection mechanism, which jointly refine the representation of small targets against complex backgrounds. Specifically, on the NUAA-SIRST and NUDT-SIRST datasets, HAFNet's IoU values exceed those of the second-best method by 2.62% and 3.52%, respectively. On the IRSTD-1K dataset, HAFNet's nIoU value is 2.87% higher than the second-best method, demonstrating its excellent generalization ability and robustness across diverse datasets.

Fig. 6 shows the ROC curves of different deep learning-based methods, further illustrating the superiority of HAFNet. The ROC curve of HAFNet consistently lies above those of competing methods, indicating a better trade-off between detection probability and false alarm rate. Notably, on NUAA-SIRST and NUDT-SIRST, HAFNet achieves the largest area under the ROC curve, reinforcing its capability to detect small infrared targets with high accuracy while maintaining the lowest false alarm rate.

Table III summarizes the experimental results on the NoisySIRST dataset. Overall, the proposed HAFNet demonstrates stable and superior performance under varying noise levels, especially under medium and high signal-to-noise ratio conditions. In most cases, HAFNet achieves the highest IoU and nIoU scores, outperforming the second-best method by 1.46% and 2.45%, respectively, under the noise setting of $\sigma_n = 10$.

TABLE III

PERFORMANCE COMPARISON OF DEEP LEARNING-BASED METHODS ON THE NOISYSIRST DATASET WITH DIFFERENT GAUSSIAN WHITE NOISES

Methods	$\sigma_n = 10$ (SNR=5.35)		$\sigma_n = 20$ (SNR=3.69)		$\sigma_n = 30$ (SNR=2.76)	
	IoU	nIoU	IoU	nIoU	IoU	nIoU
ACM	67.36	68.01	68.55	64.59	62.18	62.11
RDIAN	70.91	75.71	69.27	70.42	65.99	66.41
DNANet	77.01	74.27	70.56	68.77	68.59	62.89
UIUNet	77.77	74.74	**74.53**	72.27	68.70	**70.19**
RPCANet	65.44	67.83	50.58	55.11	44.46	48.81
MSHNet	73.54	71.21	70.91	68.40	68.77	64.12
SCTransNet	73.61	74.81	70.81	71.31	68.51	68.57
HAFNet(Ours)	**79.23**	**78.16**	72.90	**72.78**	**69.97**	69.50

F. Qualitative Results

Fig. 7 illustrates qualitative comparisons of different IRSTD methods across three benchmark datasets: NUAA-SIRST

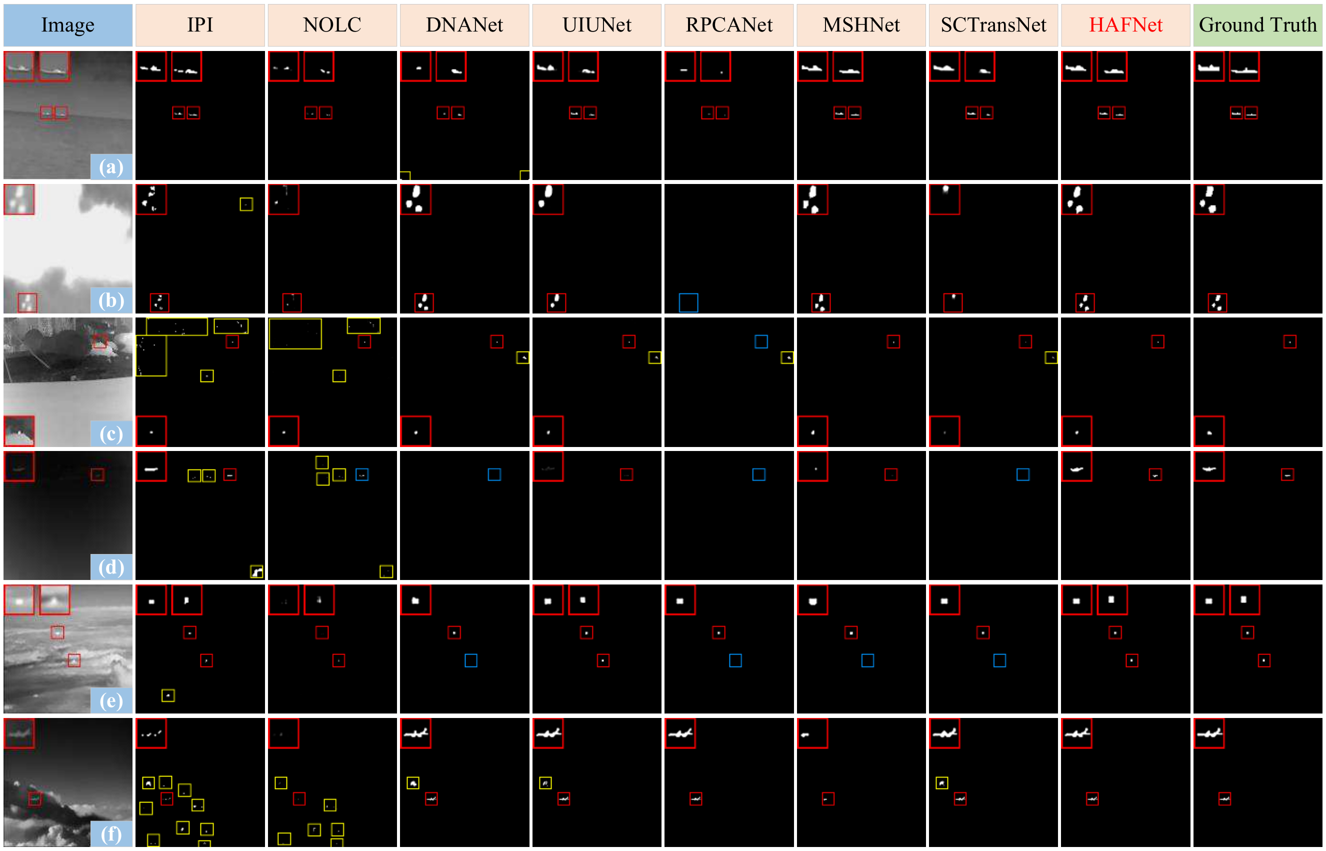

Fig. 7. Visualization of different IRSTD methods on NUAA-SIRST, IRSTD-1K, and NUDT-SIRST datasets. The red, blue, and yellow boxes represent correctly detected targets, missed detections, and false alarms, respectively. (a)–(f) Show different types of typical scene features, including sea surface, sky, and complex ground environment, to demonstrate the detection performance of the method under diverse backgrounds.

[see Fig. 7(a) and (b)], IRSTD-1K [see Fig. 7(c) and (d)], and NUDT-SIRST [see Fig. 7(e) and (f)]. Compared with traditional methods, deep learning-based methods exhibit a significant reduction in false alarms, particularly in Fig. 7(c) and (f), where traditional methods generate numerous false positives due to their reliance on handcrafted features and fixed filtering operations. The superiority of deep learning methods stems from their data-driven feature extraction, allowing them to learn target characteristics and mitigate background interference autonomously. By leveraging hierarchical feature representations, deep learning-based models more effectively differentiate real targets from complex backgrounds, leading to improved detection reliability.

Among the evaluated deep learning-based methods, the proposed HAFNet consistently achieves superior performance, surpassing existing SOTA methods in both target recall and shape reconstruction. For example, in Fig. 7(e), while DNANet, RPCANet, MSHNet, and SCTransNet exhibit missed detections, both UIUNet and HAFNet successfully detect all targets. Notably, the target shapes detected by HAFNet exhibit higher fidelity to the ground truth compared to UIUNet, underscoring the effectiveness of the proposed HFFD in preserving target morphology and enhancing segmentation precision.

The 3-D visualization results in Fig. 8 and the gradient-weighted class activation mapping (Grad-CAM) in Fig. 9 further verify the superiority of HAFNet in IRSTD. To provide an in-depth comparative analysis, we selected several representative images from the IRSTD-1K dataset for analysis. In Fig. 9, to enhance clarity and distinguish missed detections from the background, false negatives are highlighted in green, differing from the color representation used in Fig. 8. The comparison reveals that HAFNet achieves significantly more precise target localization compared to other deep learning-based methods, while effectively reducing false positives and minimizing the impact of background clutter. The Grad-CAM visualization further illustrates that HAFNet consistently focuses on true target regions with high confidence, whereas alternative methods exhibit scattered activations, leading to incomplete target detection or incorrect localization.

G. Ablation Study

1) Performance Analysis Among Each Module: To systematically evaluate the contribution of each proposed module, we conduct an ablation study using U-Net [17] as the baseline, progressively incorporating DS, DSPM, HFFE, and HFFD to analyze their individual and combined impact on detection performance. As shown in Table IV, the gradual addition of these modules leads to consistent improvements in IoU, nIoU, and F1 across three benchmark datasets. Notably, DSPM significantly enhances feature extraction, increasing IoU, nIoU, and F1 by 4.93%, 5.17%, and 2.92%, respectively, demonstrating its effectiveness in capturing small target features. Further

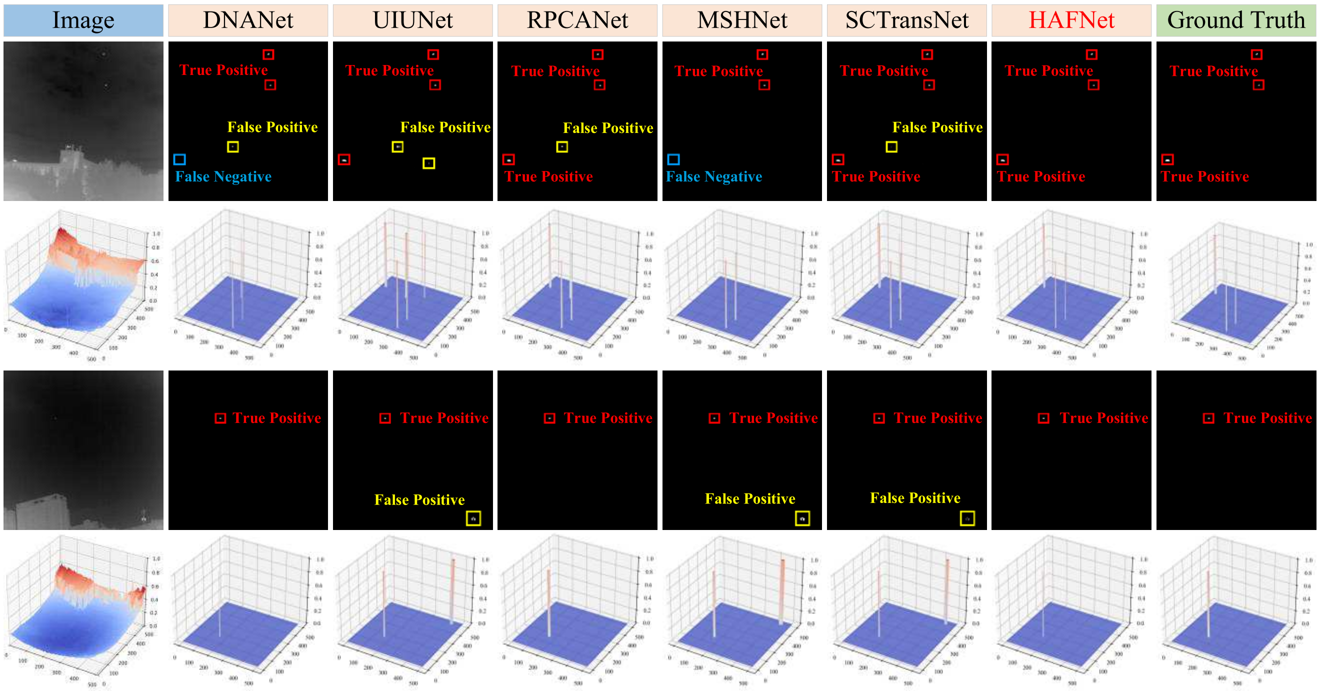

Fig. 8. Visualization of different IRSTD methods on IRSTD-1K dataset. The red, blue, and yellow boxes represent correctly detected targets, missed detections, and false alarms, respectively.

TABLE IV

ABLATION STUDY TO EVALUATE THE DSPM, DS, HFFE, AND HFFD MODULE, IN AVERAGE IoU (%), nIoU (%), AND F1 (%) ON NUAA-SIRST, IRSTD-1K, AND NUDT-SIRST

UNet	+DS	+DSPM	+HFFE	+HFFD	IoU	nIoU	F1
✓	×	×	×	×	75.02	75.94	85.47
✓	✓	×	×	×	75.17	76.04	85.56
✓	✓	✓	×	×	79.95	81.11	88.39
✓	✓	✓	✓	×	80.71	81.51	88.84
✓	✓	✓	×	✓	80.03	81.35	88.45
✓	✓	✓	✓	✓	**81.14**	**82.11**	**89.13**

TABLE V

ABLATION STUDY TO EVALUATE THE COMPONENTS OF DSPM, IN IoU (%)/nIoU (%)

Model	Datasets		
	NUAA-SIRST	IRSTD-1K	NUDT-SIRST
DSPM w/o Conv	73.20/76.65	66.51/68.26	95.53/95.35
DSPM w/o D.Conv	76.81/78.40	67.53/68.79	92.66/92.72
DSPM w/o SAM	78.16/80.10	66.30/67.66	96.22/95.69
DSPM w/o CAM	76.89/79.71	66.44/64.93	95.33/95.28
DSPM	**79.19/81.00**	**67.95/69.23**	**96.28/96.11**

integrating HFFE and HFFD enhances multiscale feature fusion and target reconstruction, yielding final improvements of 6.12%, 6.17%, and 3.66% over the baseline. These results validate the effectiveness of our hierarchical fusion strategy in improving target localization, structural integrity, and robustness against background interference.

Furthermore, the Grad-CAM visualization results for different model variants are presented in Fig. 10, providing an intuitive illustration of the impact of each module on feature representation. The baseline U-Net exhibits weak activation in the target area, leading to large-scale false detections. Following the introduction of DSPM, the feature extraction capability is improved, and the target response becomes more pronounced; however, some false detections remain due to the expansion of the receptive field. The addition of HFFE strengthens multiscale feature fusion, effectively reducing the false alarm rate. HFFD further optimizes the target reconstruction process and suppresses background interference, thereby enhancing the clarity of the target area. Finally, the complete HAFNet model achieves the most accurate target localization and substantially improves detection accuracy and robustness.

2) Performance Analysis of DSPM Internal Components: The ablation study on DSPM, shown in Table V, demonstrates the substantial contribution of each component to detection performance. Removing spatial attention (without SAM) weakens target localization, causing a drop in nIoU by 1.57% on IRSTD-1K, while eliminating channel attention (without CAM) diminishes interchannel feature correlation, leading to a 0.83% decrease in nIoU on NUDT-SIRST. Furthermore, the comparison between standard convolution (Conv) and dilated convolution (D.Conv) indicates that although dilated convolution expands the receptive field, relying solely on it compromises local feature extraction, leading to a more substantial IoU decline of 5.99% when standard convolution is removed, as opposed to a 2.38% reduction when dilated convolution is excluded on NUAA-SIRST. The complete DSPM, integrating both convolution types and attention mechanisms, achieves the highest IoU/nIoU metrics among three datasets,

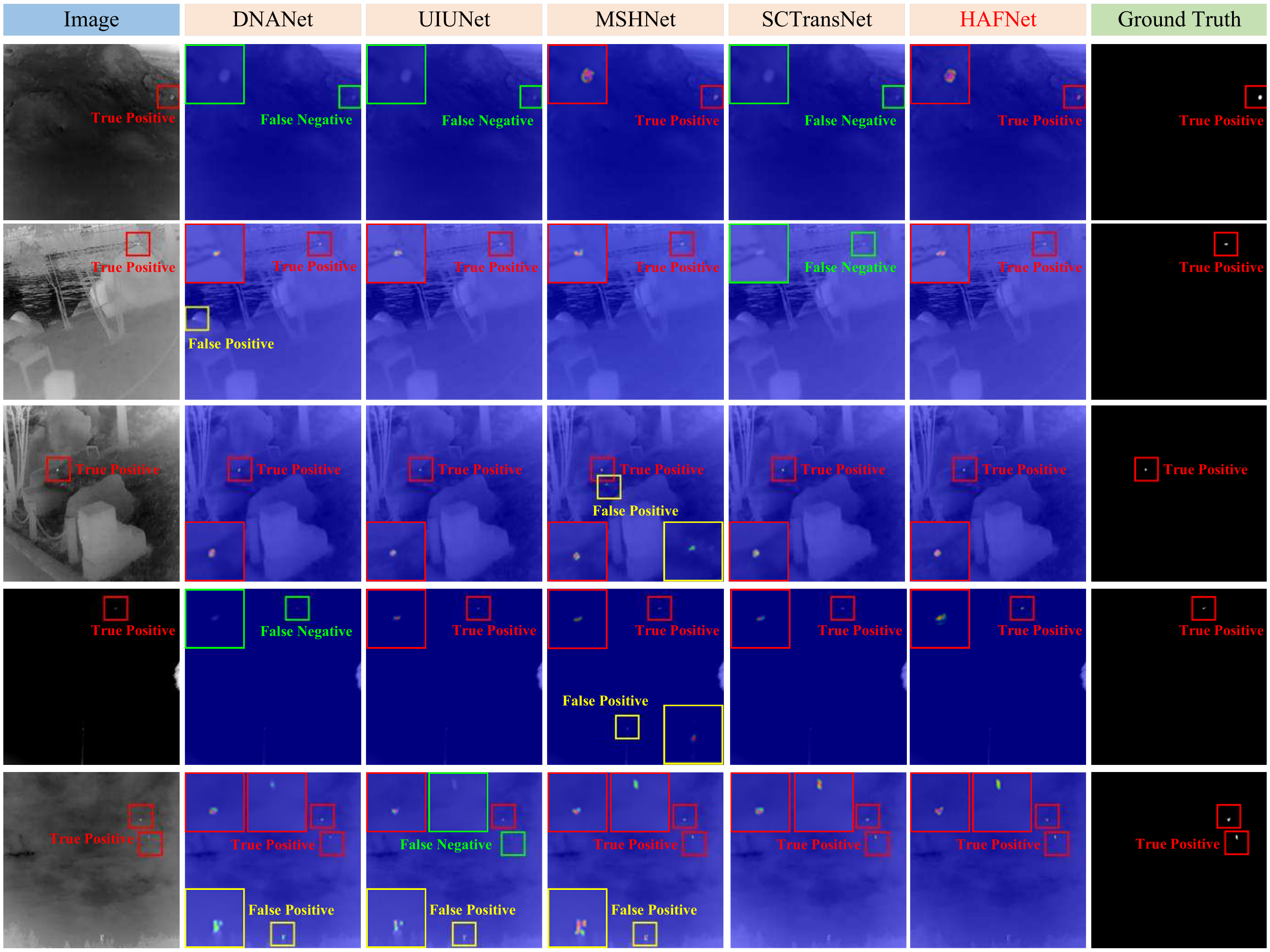

Fig. 9. Feature visualization comparison on IRSTD-1K dataset. True positive detections, false positives, and missed detections are represented by red, yellow, and green bounding boxes, respectively. Compared with other methods, the proposed HAFNet shows superior performance with higher true positive detections and fewer false positives, and missed detections.

TABLE VI
ABLATION STUDY TO EVALUATE THE COMPONENTS OF HFFE, IN IoU (%)/NIoU (%)

Model	Datasets		
	NUAA-SIRST	IRSTD-1K	NUDT-SIRST
HFFE w/o SAM	77.63/79.69	66.55/68.35	96.14/96.17
HFFE w/o CoordAtt	76.11/78.93	67.70/68.46	96.25/**96.21**
HFFE w/o $\otimes^1$	78.45/80.31	67.05/68.32	95.72/95.40
HFFE w/o $\otimes^2$	76.61/78.95	67.45/68.42	96.17/95.80
HFFE	**79.19/81.00**	**67.95/69.23**	**96.28**/96.11

validating its effectiveness in enhancing target-background discrimination and improving small target detection accuracy.

3) Performance Analysis of HFFE Internal Components: To assess the contribution of each submodule in HFFE, we conduct an ablation study by systematically removing individual components and evaluating model performance across three datasets, as summarized in Table VI. The results indicate that each component plays a crucial role in enhancing detection accuracy, as evidenced by significant performance degradation upon the removal of key elements. Specifically, removing spatial attention (without SAM) reduces nIoU by 0.88% on IRSTD-1K, weakening the model's ability to highlight target regions. Eliminating coordinate attention (without CoordAtt) leads to a 3.08% drop in IoU and a 2.07% decrease in nIoU on the NUAA-SIRST dataset, despite a slight increase in nIoU (+0.10%) on NUDT-SIRST. This suggests that, although CoordAtt does not consistently improve the detection of highly distinguishable targets, it remains essential for handling more challenging detection scenarios. In addition, removing element-wise multiplications $\otimes^1$ and $\otimes^2$, which correspond to feature recalibration in (13) and (14), as well as post-CoordAtt fusion, results in consistent performance degradation across all datasets, confirming their importance in cross layer feature interaction and refinement. Overall, the HFFE module effectively integrates multilevel feature representations, enabling fine-grained target enhancement while reducing false detections caused by background clutter.

4) Performance Analysis of HFFD Internal Components: The ablation study in Table VII evaluates the impact of each HFFD component on detection accuracy. Removing hierarchi-

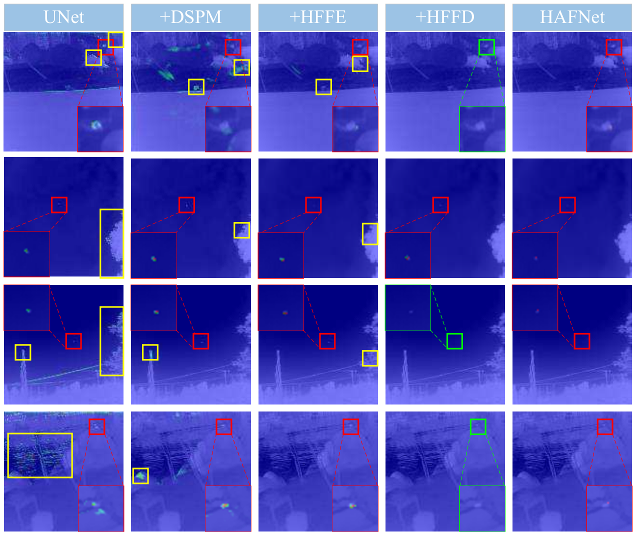

Fig. 10. Visualization of Grad-CAM for different variants in the performance analysis among each module. True positive detections, false positives, and missed detections are represented by red, yellow, and green bounding boxes, respectively.

TABLE VII

ABLATION STUDY TO EVALUATE THE COMPONENTS OF HFFD, IN IOU (%)/NIOU (%)

Model	Datasets		
	NUAA-SIRST	IRSTD-1K	NUDT-SIRST
HFFD w/o F_{en}	76.99/79.07	66.89/68.20	95.72/95.30
HFFD w/o F_{HFFE}	75.91/78.45	67.73/68.50	95.71/95.38
HFFD no FD	78.51/80.15	67.50/68.67	95.55/95.18
HFFD	**79.19/81.00**	**67.95/69.23**	**96.28/96.11**

cal semantic information (F_{HFFE}) results in the most significant performance drop, reducing IoU by 3.28% and nIoU by 2.55% on NUAA-SIRST, underscoring its critical role in target clarity and background suppression. Excluding encoder features (F_{en}) degrades structural integrity, leading to decreases in IoU and nIoU across all datasets, confirming that skip connections are crucial for fine-detailed retention and effective feature propagation. While removing feature decomposition (FD) has a smaller impact, it still reduces IoU and nIoU, indicating its contribution to multiscale feature fusion and robustness against complex backgrounds. These findings validate that HFFD optimally integrates multisource information, achieving superior small target detection across all datasets.

5) Effectiveness of DSPM Feature Extraction: As mentioned in Section I, most of the existing deep learning-based IRSTD methods are usually based on various classification backbones as feature extraction modules. In order to verify the advantages of DSPM, this section will compare two feature extraction modules in other IRSTD methods, namely the dense nested interactive module in DNANet and the residual block in ResNet (which is also a commonly used feature extraction module in many IRSTD methods). Specifically, we replace DSPM in HAFNet with these two modules and construct two variant models, named H.DNIM and H.RB. The experimental results are shown in Table VIII. Our HAFNet achieves the best

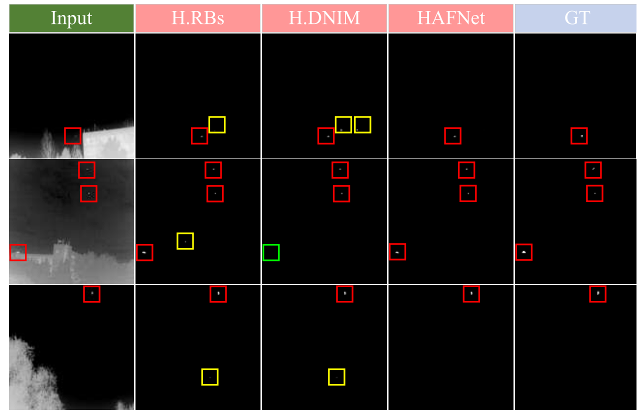

Fig. 11. Visual results obtained by different variants in the effectiveness of DSPM. The red, green, and yellow boxes represent correctly detected targets, missed detections, and false alarms, respectively.

TABLE VIII

ABLATION STUDY TO EVALUATE THE VARIANTS PRODUCED BY DIFFERENT FEATURE EXTRACTION MODULES, IN IOU (%)/NIOU (%)/F1 (%)

Model	Datasets		
	NUAA-SIRST	IRSTD-1K	NUDT-SIRST
H. DNIM	78.62/80.04/88.03	66.75/68.17/80.05	94.83/95.28/97.34
H. RBs	77.37/79.23/87.24	66.77/66.52/80.06	91.07/90.57/95.32
HAFNet	**79.19/81.00/88.39**	**67.95/69.23/80.91**	**96.28/96.11/98.10**

performance on both datasets. This result fully demonstrates the excellent performance and effectiveness of the proposed DSPM in feature extraction.

Fig. 11 shows the detection results of different model variants. From the visual results, it can be seen that H.RBs and H.DNIM have a large number of false alarms and miss detections, indicating that previous feature extraction methods are limited by fixed receptive fields and have limitations in effectively capturing small target features. In contrast, HAFNet significantly reduces false alarms while ensuring accurate target positioning, verifying the excellent feature extraction ability of DSPM.

6) Impact of the Network Depth in HAFNet: To analyze the impact of network depth on detection performance, we conducted an ablation study, with results summarized in Table IX. Following the setting of previous works [18], [25], we started with a three-layer network structure. Specifically, a three-layer U-Net model with two downsampling layers (layer 3) achieves 78.10% IoU and 80.26% nIoU on the NUAA-SIRST dataset, and 95.59% and 95.33% on the NUDT-SIRST dataset. These results show that the proposed HAFNet framework still has strong detection capabilities even in a shallow architecture, verifying its efficient feature extraction performance.

As network depth increases, performance steadily improves. For instance, layer 4 achieves 78.38%/80.35% and 96.03%/95.91% IoU/nIoU, benefiting from deeper feature representation and enhanced contextual learning. The full five-layer HAFNet model further boosts detection accuracy, achieving the best performance across both datasets. However, beyond five layers (layer 6), performance begins to decline,

TABLE IX
ABLATION STUDY TO EVALUATE THE NETWORK DEPTH OF HAFNET, IN IOU (%)/NIOU (%)

Model	Params(M)	Flops(G)	Datasets	
			NUAA-SIRST	NUDT-SIRST
Layer 3	0.81	12.47	78.10/80.26	95.59/95.33
Layer 4	3.36	18.59	78.38/80.35	96.03/95.91
Layer 6	40.64	29.52	78.19/80.77	95.58/95.81
HAFNet	13.46	24.50	**79.19/81.00**	**96.28/96.11**

TABLE X
COMPUTATIONAL COMPLEXITY AND PERFORMANCE OF COMPETING METHODS, IN PARAMS, FLOPS, AND INFERENCE LATENCY

Model	Params(M)	FLOPs(G)	Inference latency(ms)
DNANet	4.70	14.26	22.09
UIUNet	50.54	54.43	16.96
MSHNet	4.07	6.10	10.86
SCTransNet	11.19	20.24	25.48
HAFNet (ours)	13.55	24.68	24.77

likely due to excessive feature compression and increased difficulty in preserving fine-grained target details. To maintain a balance between model complexity and detection accuracy, we adopt a five-layer U-Net architecture as the optimal design for HAFNet.

H. Model Complexity Analysis

As shown in Table X, a comparative analysis is conducted of several mainstream IRSTD methods across three key dimensions: model parameters, floating point operations (FLOPs), and inference latency. MSHNet and DNA-Net employ a small number of parameters and low FLOPs, indicating high computational efficiency; however, their detection performance remains relatively limited. While UIUNet exhibits excellent detection accuracy, albeit with significantly greater computational complexity and a larger number of parameters, it thereby limits its suitability for practical deployment. In contrast, the proposed HAFNet strikes a favorable balance between accuracy and efficiency. With 13.55-M parameters, 24.68-G FLOPs for a 256×256 input, and an inference latency of 24.77 ms on a GPU, HAFNet achieves SOTA performance while maintaining moderate computational demands, thereby offering greater practicality for real-world detection scenarios.

V. CONCLUSION

In this article, a novel network architecture is proposed for the IRSTD task, namely HAFNet, which achieves dual improvements on the original U-Net architecture. Regarding the feature extraction backbone, DSPM is designed to combine standard and dilated convolutions with SAM/CAM, significantly improving the feature extraction capability of the network and effectively separate the target from the background. Furthermore, we further expand the skip connections in traditional U-Net by introducing hierarchical feature fusion modules (HFFE and HFFD) at both ends of the encoder and decoder, thereby constructing hierarchical skip connections. This improvement not only enhances the information interaction efficiency between the encoder and decoder, but also significantly strengthens the decoder's ability to restore the target. Comprehensive evaluations on multiple public datasets demonstrate the effectiveness and superiority of HAFNet in the IRSTD task. Despite its strong performance, HAFNet has a limitation: the dual-branch feature extraction and layer-wise fusion modules introduce a relatively high number of parameters and computational complexity, which may hinder its real-time deployment on resource-constrained platforms. In future work, we will focus on lightweight model design and inference optimization to enhance the deployability of HAFNet in practical applications.

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